

AI Projects, Concepts, and Prototypes

Includes Forge Assist, the early prototype that evolved into Honeywell's first customer-facing AI assistant, now deployed as Maintenance Assist.

Vertex Assist

This is a concept for a high tech gas detector that is placed in a semiconductor FAB to prevent any issues during the making of silicon chips in facilities managed by Samsung, Intel and others. I created the touch screen application and presented these concepts that would use generative AI to provide recommendations and insights based on data from the gas analyzers including alarms and faults in the optics, pump, filter or chemcassette.

Key Features (Product Outcomes)

- Generative AI Technology – Leverages Honeywell's proprietary data and documentation for step-by-step maintenance and repair guidance.
- Natural Language Processing – Understands user queries in plain language and provides context-aware responses.
- Conversational Interface – Simplifies complex maintenance tasks through an intuitive chat-style interface.
- Extensive Knowledge Base – Connects to Honeywell's ecosystem of manuals, service tickets, and knowledge articles.
- Interactive, User-Friendly Design – Built for both desktop and mobile, requiring minimal user training.

Welcome to Vertex Assist

Expert help from historical data, tickets, studies and manuals.
Just ask ...

Troubleshoot issues

Ask questions about specific tasks

Maintenance schedule?

When do I need to replace a filter on AZ1?

Technical support?

Configuration and settings help

General questions?

Expert solutions without a call

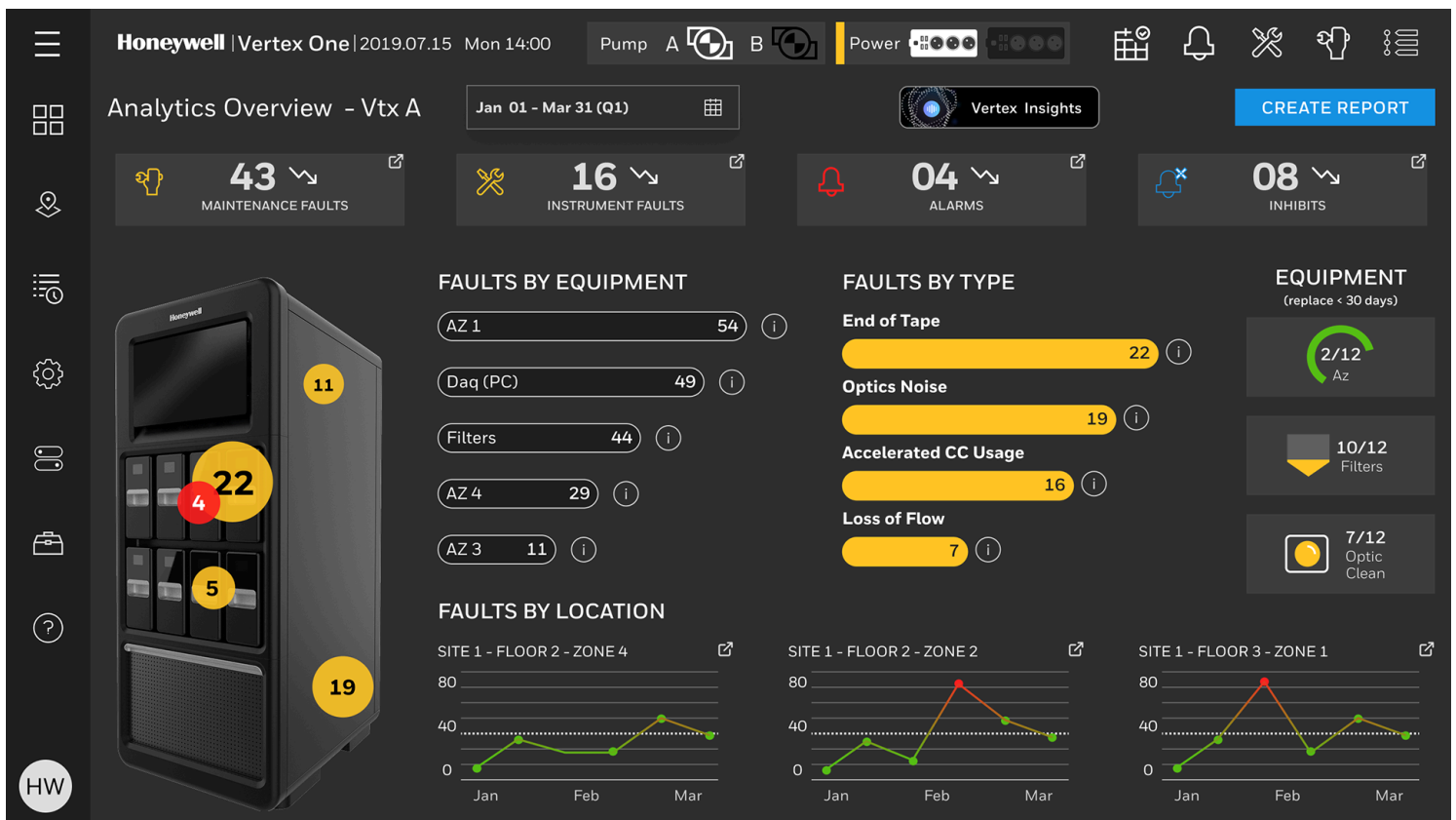
Enter a query such as "How do I replace a filter"



ged filter

tex Assist

CANCEL

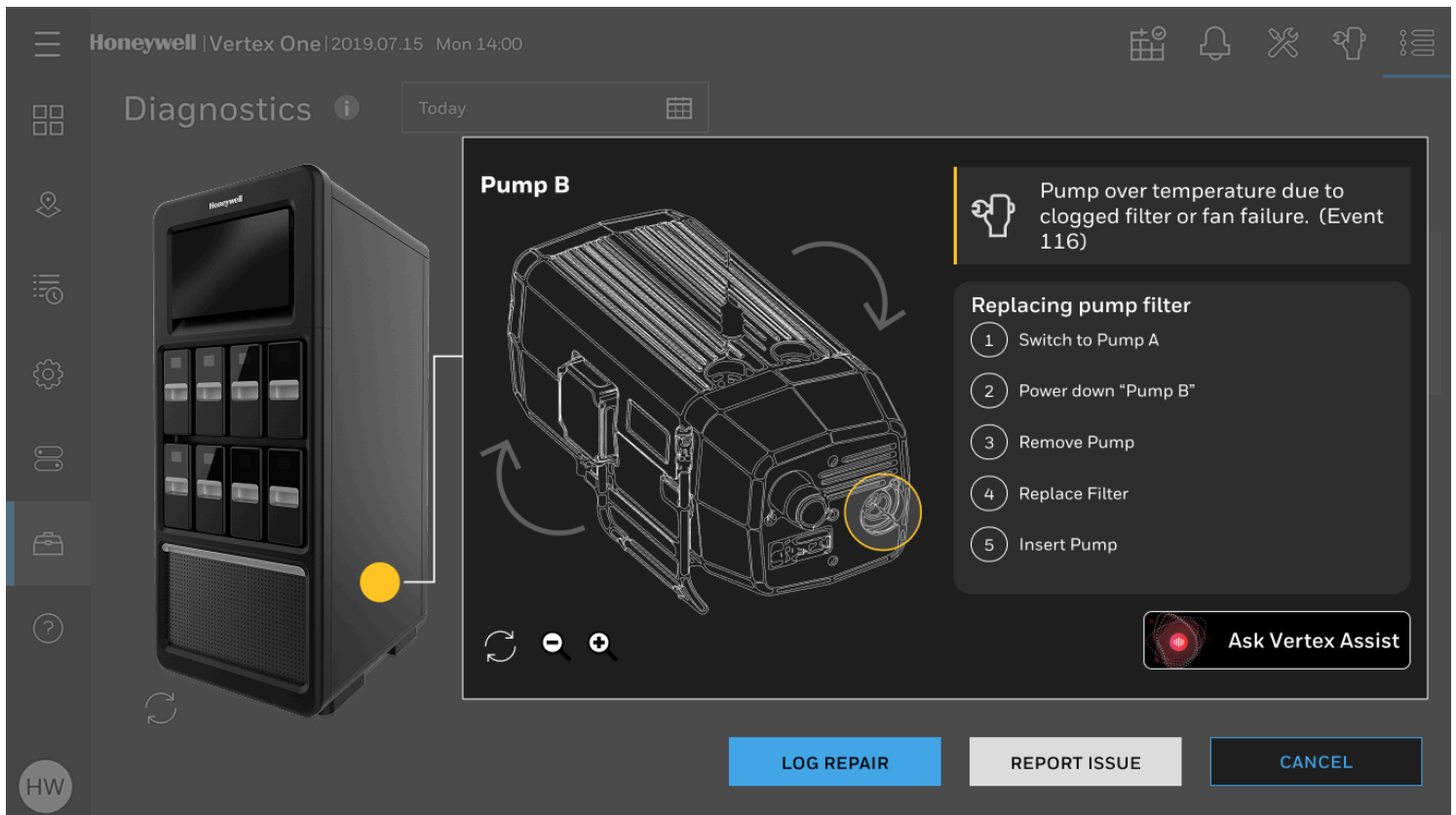


Vertex Digital Twin (AI-generated guided steps for maintenance)

This is the diagnostics application for Vertex One which is Honeywell's next generation Vertex machine used in semiconductor FABs. The system would use AI to generate custom maintenance instructions for

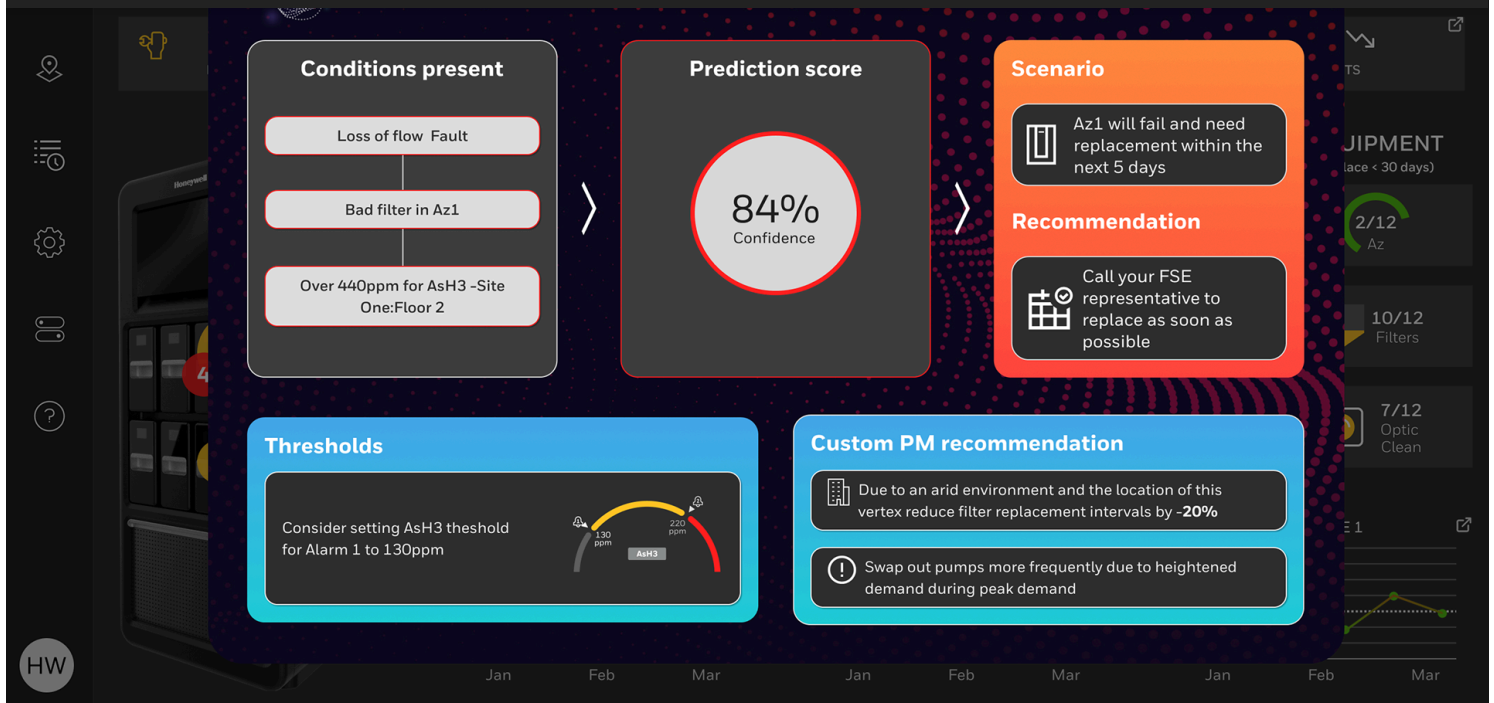
Simplifies troubleshooting, speeding up repair times and decision making.

- Cost Savings - Streamlines maintenance workflows, reducing operational overhead.
- Enhanced Productivity - Provides actionable guidance that boosts technician effectiveness.
- Training Efficiency - accelerates onboarding and continuous learning for new technicians.
- Reduced Downtime - Enables predictive servicing and faster issue resolution.



Vertex Predictive Insights and Recommendations

Applies AI and machine learning to forecast potential equipment issues before they occur. The system generates preventive maintenance recommendations with a confidence score to indicate prediction reliability. By providing early warnings and guided corrective actions, the application helps prevent costly downtime in semiconductor FAB environments, improving overall equipment efficiency and reliability.



AI-generated Insights and Widgets

I designed the "AI-Generated Insights and Widgets" dashboard which includes customized data visualizations and enables users to create AI-generated widgets for specific insights into issues affecting performance and safety. The platform acts as a central hub for large industrial factories as well as first responders to actively manage safety and integrates data from connected safety sensors, gas detection devices, and connected workers. The AI model would bubble up trends, predict risks, and recommend preventive actions before dangerous incidents occur.

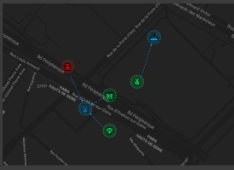
- Designed the next-generation worker-safety ecosystem integrating connected gas detectors, fleet management, and predictive analytics.
- Mapped workflows and information architecture for device setup, check-in/out, and compliance reporting
- Co-created AI-driven dashboards for Opintel and safety outcomes.
- Integrated a "Fleet Copilot", a conversational AI assistant that delivers insights ("Which workers had the most alarms today?") and generates automated reports.
- Directed UX research across global partners and competitor analysis
- Helped build design system updates in React + Figma, including accessibility and mobile responsiveness.



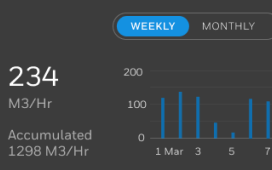
ACTIVE



CONNECTED DEVICES



FLOW RATE



WORKER PERFORMANCE

Injuries

Trend indicator

234

Non-life threatening

Show details

SUSTAINABILITY

CO2

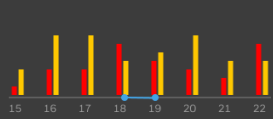
Trend indicator

44

Show details

Meter Management

ALARMS AND ALERTS



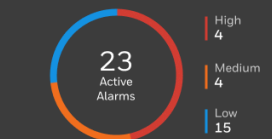
Tank Management

STORAGE



Alarm Management

ACTIVE



Insight Generator

Data Viz Library

Enter a query such as "Show how many alerts Refinery ..."



CONNECTED ASSETS



Apps

Search



Bad weather is approaching Houston and there are alarms occurring at refinery ABC

Houston, USA

25 mins ago

Take Action

AI RECOMMENDATIONS



Worker John Smith training not complete and currently in unsafe conditions

Lincolnshire, USA

25 mins ago

Take Action



Refinery ABC contributed the most to CO2 emissions last month and should sensor XYZ activated the most alarms

Houston, USA

45 mins ago

Take Action



Devices

BUMP AND CALIBRATION



Alarms

ACTIVE

12 ALARMS

12 ALERTS

February

15 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 01 02 03 04 05 06 07 08 09 10 11 12 13 14

Assets

ACTIVE





AI Assistant (Maintenance Assist)

Maintenance Assist is Honeywell's advanced Generative AI digital assistant, designed to simplify complex systems and empower personnel to maintain, troubleshoot, and repair critical assets easily and efficiently.

As part of Honeywell's Enterprise Worker & Asset Safety (EWAS) initiative, my research and design prototypes directly informed the product's conversational UX, context models, and information architecture. I collaborated with engineers to explore LLM integrations, data pipelines, and React-based UI components that would later define Maintenance Assist's core experience.

Key Contributions

- **AI Conversational UX** – Prototyped the natural language assistant that guides maintenance technicians step-by-step through troubleshooting and repair workflows.
- **Information Architecture + Interaction Design** – Defined how contextual insights (alarms, manuals, diagnostics) appear dynamically in conversation, reducing cognitive load.
- **Design System Integration** – Collaborated with engineers to implement the conversational assistant within Honeywell's React + Figma design system, ensuring accessibility (WCAG 2.1).
- **User Validation** – Conducted field VOCs and usability sessions with maintenance personnel and partners (KD Analytics, Virgin Media, AllSafe) to evaluate comprehension, efficiency, and trust in AI outputs.
- **Cross-Product Alignment** – Extended the AI assistant framework to other Honeywell digital products like Fleet Copilot and Operations Assist for predictive and preventive maintenance.

The below concept is using maintenance assist application on a tablet which is the same as the mobile app we viewed earlier.



I have compiled steps to perform the installation.

[Guide Me](#)[Download Manual](#)

Was this helpful? [Yes](#) [No](#)

Data Source:

Guided by



Here are the 5 steps to install motorized drive rollers on conveyor #2322. The estimated time is 40 minutes to complete installation.

- Remove drive roller's connector
Estimated time: 10 minutes
- Loosen both nuts on the roller bracket and unclamp the shaft.
Estimated time: 5 minutes
- Remove completely other end of the roller
Estimated time: 8 minutes
- Install replacement and plug in connector
Estimated time: 10 minutes
- Reconnect and turn on the power
Estimated time: 2 minutes
- Provide installation evidence
Estimated time: 5 minutes



Speak This?

Was this helpful? [Yes](#) [No](#)

Data Source: NexasNeural

I need help with the plug in connector



Here is a video that can help



Thanks, I'm



Suggested next steps:

Create Note

Copy Link

New Chat

Honeywell's Forge Assist (AI Assistant)

This prototype demonstrates a dual-AI interaction model where both the user and the assistant leverage AI. Built using ProtoPie's voice trigger method, it enables a warehouse maintenance technician to initiate tasks hands-free. The AI voice assistant provides guided repair instructions for a conveyor system, while the phone's camera and computer vision detect mechanical issues in real time and offer context-sensitive assistance.

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